Ata B. Karagoz, Ph.D.

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Appointments

The University of Chicago, Chicago, IL — Aug. 2025 – Present
Postdoctoral Scholar, Institute for Mind and Biology (Bakkour Lab)

Education

Washington University in St. Louis, St. Louis, MO — Aug. 2020 – Aug. 2025
Ph.D. in Psychological and Brain Sciences
Dissertation: Segmentation and Memory in Interactive, Inferential Events
Advisor: Zachariah M. Reagh, Ph.D.

Washington University in St. Louis, St. Louis, MO — Aug. 2020 – Dec. 2022
M.A. in Psychological and Brain Sciences

The University of Texas at Austin, Austin, TX — Aug. 2014 – May 2018
B.S. in Neuroscience

Research Interests

How people learn about the world and use what they have learned to guide value-based decisions, studied through the interaction of memory and decision-making and formalized with computational models. Related interests include event memory, reinforcement learning, naturalistic stimuli, and functional neuroimaging.

Fellowships, Grants, and Awards

Ruth L. Kirschstein NRSA Postdoctoral Fellowship (F32) — 2026
Sponsor: Akram Bakkour, Ph.D. | Scored: Impact 30, 20th percentile (pending funding decision)

Complexity-GAINs International Summer School, Santa Fe Institute — Aug. 2023
Competitive selection (∼30 participants); NSF-funded; Isaac Newton Institute, Cambridge, UK

- One of three winning final-project teams.
- The win led to a follow-up workshop at SFI (see Service).

NINDS Interdisciplinary Training Grant (CCSN) — July 2022 – July 2024
Cognitive, Computational, and Systems Neuroscience; funded by NINDS and the McDonnell Center for Systems Neuroscience

Peer-Reviewed Publications

Tang, R., **Karagoz, A.B.**, Delarazan, A.I., & Reagh, Z.M. (2026). Active linking through narratives facilitates associative inference. *Proceedings of the National Academy of Sciences.*

Karagoz, A.B., Kool, W.†, & Reagh, Z.M.† (2025). Free recall is shaped by inference and scaffolded by event structure. *Communications Psychology.*

Karagoz, A.B., Moran, E.K., Barch, D.M., Kool, W.†, & Reagh, Z.M.† (2025). Evidence for shallow cognitive maps in schizophrenia. *Cognitive, Affective, & Behavioral Neuroscience*.

Karagoz, A.B., Reagh, Z.M.†, & Kool, W.† (2024). Construction and use of cognitive maps in model-based control. *Journal of Experimental Psychology: General*.

Karagoz, A.B., Morse, S.J., & Reagh, Z.M. (2023). Cortico-hippocampal networks carry information about characters and their relationships in an extended narrative. *Neuropsychologia*.

Sherrill, K.R., Molitor, R.J., **Karagoz, A.B.**, Atyam, M., Mack, M.L., & Preston, A.R. (2023). Generalization of cognitive maps across space and time. *Cerebral Cortex*.

† denotes equal contribution.

Preprints

Karagoz, A.B., & Bakkour, A. (2026). Learning and using value maps in multi-dimensional choice. Preprint: https://osf.io/preprints/psyarxiv/2tpuv_v1.

Bouffard, N.R., Delarazan, A.I., **Karagoz, A.B.**, Zacks, J.M., & Reagh, Z.M. (2026). Age differences in hippocampal neural timescales during movie-viewing. Preprint: https://doi.org/10.64898/2026.05.05.723065.

Morse, S.J., **Karagoz, A.B.**, & Reagh, Z.M. (2023). Event boundaries directionally influence item-level recognition memory. Preprint: https://psyarxiv.com/h8bj2/.

Manuscripts in Preparation

Karagoz, A.B., Morse, S.J., Reagh, Z.M.†, & Kool, W.† (in prep.). Complementary roles for PHC and lateral OFC in cognitive map generation.

Karagoz, A.B., Kool, W.†, & Reagh, Z.M.† (in prep.). Uncertainty over event extent and context shapes memory in interactive events.

Talks & Symposia

Karagoz, A.B.*, Kool, W., & Reagh, Z.M. (2024). Event boundaries and interactive inference shape the content and organization of memory. Nanosymposium talk, Society for Neuroscience (SfN), Chicago, IL.

Karagoz, A.B.*, Kool, W., & Reagh, Z.M. (2024). Representations of cognitive maps underlying model-based planning. Symposium talk, Cognitive Neuroscience Society (CNS).

Karagoz, A.B.*, Kool, W., & Reagh, Z.M. (2023). Cognitive and neural representations of cognitive maps underlying model-based planning. Nanosymposium talk, Society for Neuroscience (SfN).

Poster & Co-Authored Presentations

Karagoz, A.B.*, & Bakkour, A. (2026). Tracking attribute sampling during categorical and relational value inference. Poster, Cognitive Neuroscience Society (CNS).

Karagoz, A.B.*, Kool, W., & Reagh, Z.M. (2023). The influence of rule-based event boundaries on item memory. Poster, International Conference on Learning and Memory (LearnMem).

Morse, S.J.*, **Karagoz, A.B.**, & Reagh, Z.M. (2023). Event boundaries at encoding influence mnemonic discrimination. Poster, International Conference on Learning and Memory (LearnMem).

Roome, H.E.*, Sherrill, K.R., Nguyen, K.V., **Karagoz, A.B.**, Coughlin, C.A., & Preston, A.R. (2022). Medial temporal lobe error signals mediate developmental differences in spatial memory precision. Nanosymposium talk, Society for Neuroscience (SfN).

Sherrill, K.R.*, Roome, H.E., **Karagoz, A.B.**, Long, J.M., & Preston, A.R. (2022). Emergence of hippocampal and ventromedial prefrontal cortex context-dependent coding during virtual navigation. Nanosymposium talk, Society for Neuroscience (SfN).

Karagoz, A.B.*, Reagh, Z.M., & Kool, W. (2022). Constructing and using cognitive maps for model-based control. Poster, Reinforcement Learning and Decision Making (RLDM).

Karagoz, A.B.*, & Reagh, Z.M. (2022). Representations of perceptual versus semantic relationships among characters in naturalistic events. Poster, Context and Episodic Memory Symposium (CEMS).

Reagh, Z.M.*, Morse, S.J., Fishman, R., Angulo-Lopera, S., **Karagoz, A.B.**, & Delarazan, A.I. (2022). Event boundaries at encoding influence mnemonic discrimination. Poster, Cognitive Neuroscience Society (CNS).

Delarazan, A.I.*, **Karagoz, A.B.***, Montchal, M.E., Yassa, M.A., Ranganath, C., & Reagh, Z.M. (2021). Hippocampal and entorhinal contributions to naturalistic event context reinstatement. Virtual poster, Society for Neuroscience (SfN).

Karagoz, A.B.*, Reagh, Z.M., & Kool, W. (2021). The construction and use of cognitive maps in model-based control. Virtual poster, Psychonomic Society Annual Meeting.

Karagoz, A.B.*, & Reagh, Z.M. (2021). Decoding perceptual and semantic relatedness among characters in naturalistic events. Virtual poster, Cognitive Neuroscience Society (CNS).

* *denotes presenting author.*

Research Experience

Bakkour Lab, The University of Chicago — Aug. 2025 – Present
Postdoctoral Scholar / Supervisor: Akram Bakkour, Ph.D.

- Design behavioral and fMRI studies of memory-guided, value-based decision-making.
- Collect and analyze fMRI datasets.
- Develop computational models of choice behavior.

Complex Memory Lab, Washington University in St. Louis — Aug. 2020 – Aug. 2025
Graduate Research Assistant / Supervisor: Zachariah M. Reagh, Ph.D.

- Designed and ran behavioral and fMRI studies of event memory and model-based control.
- Built computational models linking event structure, inference, and recall.

Preston Lab, The University of Texas at Austin — Jan. 2018 – Aug. 2020
Lab Manager / Supervisor: Alison Preston, Ph.D.

- Coordinated data collection and analysis across memory-integration and navigation projects.
- Scheduled sessions and built the participant pool for behavioral and scanning studies.

Preston Lab, The University of Texas at Austin — Nov. 2015 – Jan. 2018
Undergraduate Research Assistant / Supervisor: Alison Preston, Ph.D.

- Assisted in fMRI scanning studies.
- Wrote MATLAB analysis scripts for ongoing projects.
- Designed a study and developed its stimulus-presentation code in MATLAB.

Teaching & Mentoring

Graduate Teaching Assistant, *Washington University in St. Louis* Jan. 2023 – May 2023
Course: Computational Cognitive Science (cross-listed, graduate and upper-level undergraduate)

- Wrote and graded Python programming assignments.
- Led weekly review sessions.
- Guided students through their final computational modeling projects.

Teaching Assistant, *Washington University in St. Louis* Jan. 2022 – May 2022
Course: Neurobiology of Learning and Memory

- Delivered a guest lecture on spatial navigation.
- Graded assignments and led exam review sessions.

Teaching Assistant, *Washington University in St. Louis* Sep. 2021 – Dec. 2021
Course: Human Learning and Memory

- Delivered a guest lecture on spatial cognition.
- Wrote and graded quizzes.
- Led exam review sessions.

Mentoring — Master's Students Oct. 2025 – Present

- Vanessa Wu (The University of Chicago), current — project on confidence at encoding and retrieval in associative inference, measured with ratings and mouse-tracking.
- Jalal Sadek (The University of Chicago), Oct. 2025 – June 2026 — project on valence framing and memory-based heuristics in value-guided choice. Now Ph.D. student in Neuroscience, Penn State.

Mentoring — Undergraduate Research Assistants June 2018 – Present
Complex Memory Lab, Washington University in St. Louis

- Ron Fishman, Jacob Tartakovsky, Sofia Angulo-Lopera (now Ph.D. student, Neuroscience, WashU)

Preston Lab, The University of Texas at Austin

- Jaida Long (now Ph.D. student, Georgia Tech), Doru Gucer (now Internal Medicine resident, Columbia), Katherine Vasquez

Service & Outreach

Co-Organizer, "Assessing Representation in Minds and Artificial Systems" Sep. 2024
3-day workshop, Santa Fe Institute / Co-organized with David Krakauer (SFI)

- Awarded to the three winning project teams from the 2023 Complexity-GAINs summer school.
- Co-presented the introductory talk for the session "Mapping the Space of Representations."
- Invited speakers: Erica Cartmill, Ben Falandays, Jacob Foster, Martha Lewis, Brice Ménard, Alessandro Palmarini, Andrew Stier, Joey Velez-Ginorio.

Ad-Hoc Peer Reviewer 2022 – Present

- *Neuron* (co-review), *Nature Reviews Neuroscience* (co-review), *Nature Communications*, *eLife* (co-review), *Cerebral Cortex*, *Cognition*, *Communications Psychology*, *Memory & Cognition*

Psychology Outreach with Elementary Schoolers Sep. 2020 – Dec. 2020

- Taught introductory psychology over Zoom.
- Designed remote perception demonstrations.

GitHub Clinic, *Complex Memory Lab* Sep. 2020

- Ran a lab-wide workshop on version control and reproducible git practice.

Professional Memberships

- Cognitive Neuroscience Society, Postdoctoral Member
- Psychonomic Society, Member

Skills

- **Programming:** Python, MATLAB, bash scripting
- **Computational Modeling:** reinforcement learning and model-based control models, hierarchical Bayesian parameter estimation, PyTorch, JAX
- **fMRI Preprocessing & Analysis:** fMRIPrep, FSL/FEAT, ANTs, multivariate pattern analysis
- **Reproducible Workflows:** git/GitHub, Jupyter, Docker/Singularity
- **High-Performance Computing:** Texas Advanced Computing Center, WashU Center for High Performance Computing, UChicago Research Computing Center
- **Languages:** English (native), Turkish (native), German (basic)